# DoubleMLDeep: Estimation of Causal Effects with Multimodal Data

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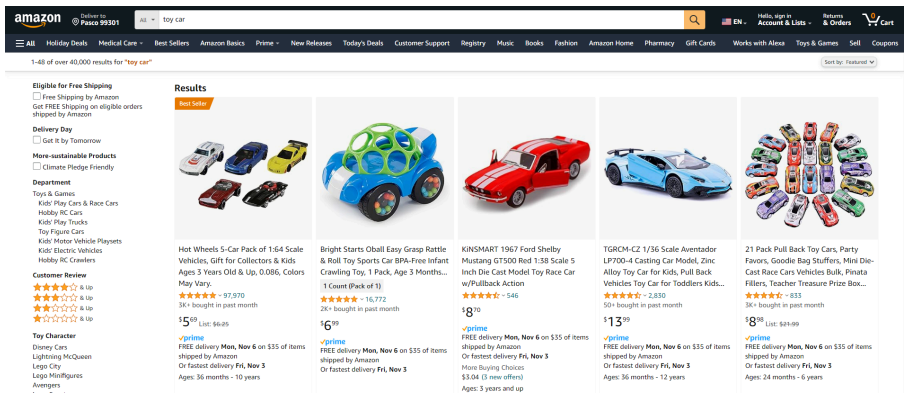
# Motivation

# Motivation

- Causal Inference mainly relies on tabular data

| | A | B | C | D | E | F | G | H | I | J | K |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | X1 | X2 | X3 | X4 | X5 | X6 | X7 | X8 | X9 | X10 | Y |
| 2 | 0,26222194 | 0,73438599 | 0,44667203 | 0,97138377 | 0,081064 | 0,28441456 | 0,30204598 | 0,42916989 | 0,45202869 | 0,48458485 | 0,69688241 |
| 3 | 0,02086748 | 0,76377164 | 0,82984213 | 0,44114913 | 0,6791843 | 0,10440879 | 0,06871173 | 0,65049866 | 0,38145381 | 0,37883627 | 0,68724919 |
| 4 | 0,80214042 | 0,60108851 | 0,3924631 | 0,64680441 | 0,52447716 | 0,71441139 | 0,92058036 | 0,50487157 | 0,0639613 | 0,72018174 | 0,70586344 |
| 5 | 0,69860016 | 0,48544746 | 0,02697306 | 0,56612716 | 0,64030572 | 0,38122836 | 0,77357736 | 0,7665354 | 0,56153968 | 0,40036439 | 0,96259876 |
| 6 | 0,76871281 | 0,52693951 | 0,23152259 | 0,99343286 | 0,19512117 | 0,52281687 | 0,74631891 | 0,51990814 | 0,8916446 | 0,13854693 | 0,15014547 |
| 7 | 0,80559982 | 0,49248903 | 0,52094313 | 0,83491804 | 0,19224862 | 0,51690107 | 0,99896405 | 0,21315395 | 0,13844972 | 0,5972433 | 0,30108879 |
| 8 | 0,14274622 | 0,68082916 | 0,21375762 | 0,12917741 | 0,101114 | 0,35962408 | 0,37101036 | 0,88481715 | 0,0917219 | 0,81144147 | 0,23748577 |
| 9 | 0,5565959 | 0,27805181 | 0,38191987 | 0,96746964 | 0,88262331 | 0,64204409 | 0,61391038 | 0,14798767 | 0,75991882 | 0,71364563 | 0,26831957 |
| 10 | 0,86680082 | 0,98940603 | 0,08032276 | 0,51284266 | 0,85067788 | 0,31879029 | 0,14212956 | 0,58889047 | 0,85003262 | 0,40984155 | 0,90320049 |
| 11 | 0,49294831 | 0,9930191 | 0,32376346 | 0,76486908 | 0,71150197 | 0,49307446 | 0,00732655 | 0,1366301 | 0,95413681 | 0,8192993 | 0,41659136 |
| 12 | 0,83367503 | 0,99203204 | 0,94500561 | 0,38442651 | 0,08267135 | 0,44319059 | 0,50597368 | 0,33789173 | 0,71150674 | 0,71723848 | 0,82498573 |
| 13 | 0,30901589 | 0,45442227 | 0,39182182 | 0,94058761 | 0,84662039 | 0,13237356 | 0,39203009 | 0,23863556 | 0,66904622 | 0,66739594 | 0,05703788 |
| 14 | 0,84854041 | 0,07220964 | 0,60564451 | 0,11235493 | 0,11423158 | 0,19425556 | 0,10525554 | 0,69400453 | 0,24273427 | 0,74521656 | 0,1934563 |
| 15 | 0,41204349 | 0,37890159 | 0,21172493 | 0,47214042 | 0,28972671 | 0,28045945 | 0,26144942 | 0,50182126 | 0,56848988 | 0,34171099 | 0,55046509 |
| 16 | 0,8270131 | 0,7550563 | 0,68080658 | 0,99718432 | 0,54748336 | 0,07330951 | 0,96389321 | 0,43492645 | 0,32444209 | 0,37929049 | 0,54724228 |
| 17 | 0,65968289 | 0,42809072 | 0,50651859 | 0,85136778 | 0,67725248 | 0,80776288 | 0,80154682 | 0,41961695 | 0,03794024 | 0,10550978 | 0,18250098 |
| 18 | 0,00250478 | 0,10976101 | 0,58665193 | 0,59539311 | 0,06932994 | 0,44532097 | 0,38697059 | 0,86107306 | 0,32445673 | 0,03897245 | 0,4593923 |
| 19 | 0,1646558 | 0,11563336 | 0,54131562 | 0,70212986 | 0,78935386 | 0,65759136 | 0,96083497 | 0,18585388 | 0,02245136 | 0,08388247 | 0,9684134 |

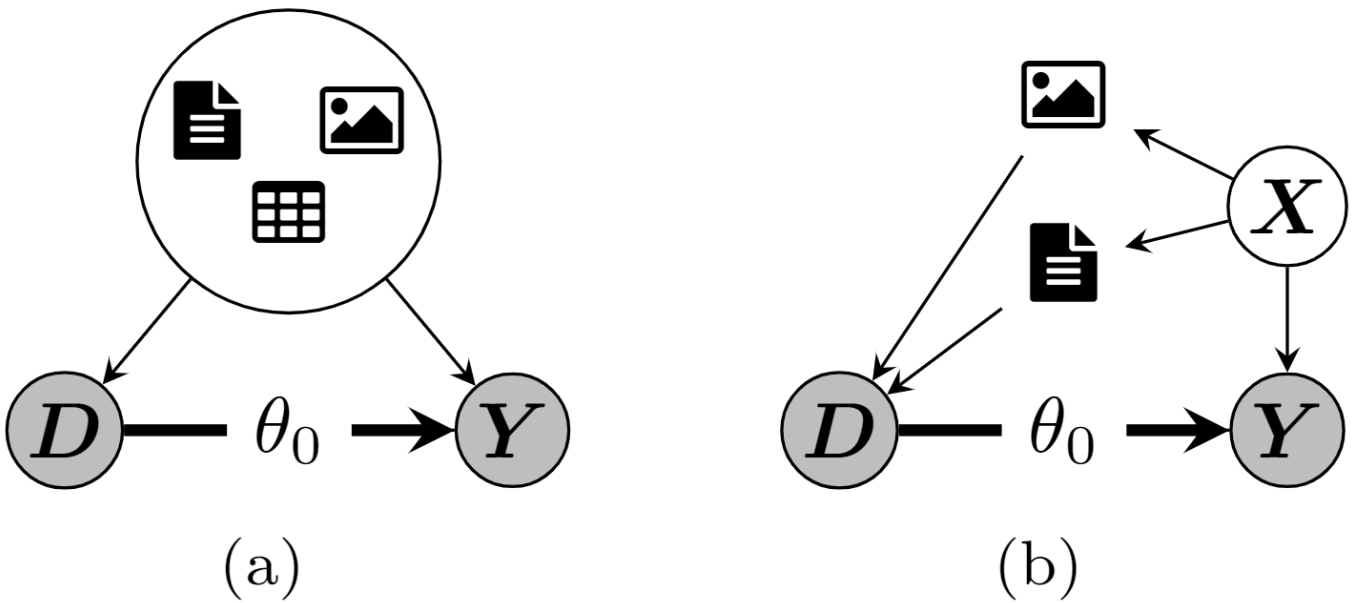
- In a lot of applications additional unstructured data is available




- We consider multimodal data as **confounders**
- Applications in Marketing, Medicine / Health, Finance, ….
  - Price elasticity of demand
  - Estimation of treatment effects conditioning on X-ray images

# Motivation

- Use multimodal data (text and images) *additionally* to conventional tabular features




Examples of directed acyclic graphs (DAGs) with image and text confounding. (a) Direct confounding via image, text and tabular data. (b) Treatment decision is driven by text and images. All backdoor paths are blocked by conditioning on both image and text data.

$\Rightarrow$ Adjust for confounding encoded in unstructured data

# Motivation

## Opportunities

- Exploit state-of-the art architectures for causal estimation (for example LLMs)
- Example: Demand estimation, account for description, images, reviews
- Huge potential for future research

## Challenges

- Recent innovations in deep learning for prediction
- Limitations of ML-based estimation due to regularization in causal estimation
- Need for effective Deep Learning architecture
- Challenges in identifying causal effects even in idealized simulations

## Our approach

- Integrate state-of-the art Deep Learning tools for estimation of Causal Parameters
- *DoubleML*[1]: Use DML to account for ML regularization bias
- Combination: *DoubleML* + Deep Learning = **DoubleMLDeep**

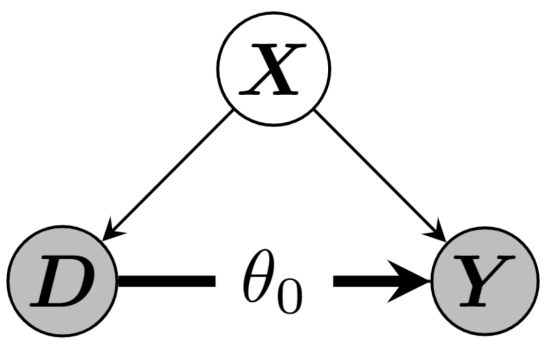
# Motivating Example - PLR

## Partially linear regression model (PLR)

$$
\begin{aligned}
Y &= \theta_0 D + g_0(X) + \varepsilon, & \mathbb{E}[\varepsilon | X, D] &= 0 \\
D &= m_0(X) + \vartheta, & \mathbb{E}[\vartheta | X] &= 0
\end{aligned}
$$

with

- $Y$ - outcome variable
- $D$ - policy/treatment variable
- $X = (X_1, \ldots X_p)^T$ - vector of additional controls
- $\varepsilon, \vartheta$ - stochastic errors




DAG for PLR Model

# Double/Debiased Machine Learning (DML)

- Use machine learning methods to fit

$$l_0(X) := \mathbb{E}[Y|X]$$
$$m_0(X) := \mathbb{E}[D|X]$$

- Construct the orthogonalized score (either cross-fitting or split the sample) for a fitted nuisance learner $\hat{\eta} := (\hat{l}, \hat{m})$

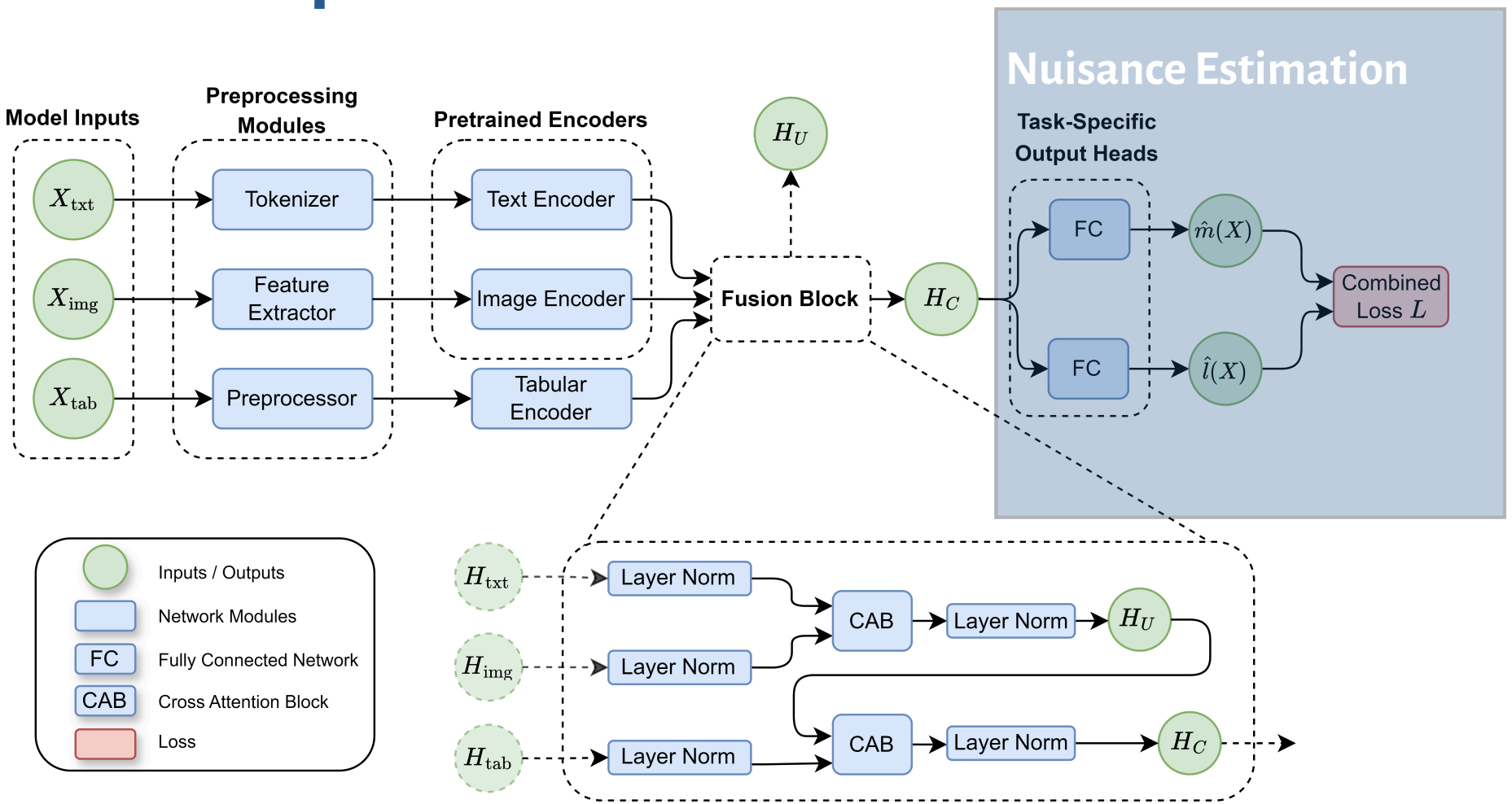
$$\psi(W, \theta, \hat{\eta}) := \Big(Y - \hat{l}(X) - \theta\big(D - \hat{m}(X)\big)\Big)\Big(D - \hat{m}(X)\Big)$$

- Compute the estimate as the solution of

$$0 = \frac{1}{n}\sum_{i=1}^{n} \psi(W, \hat{\theta}, \hat{\eta})$$

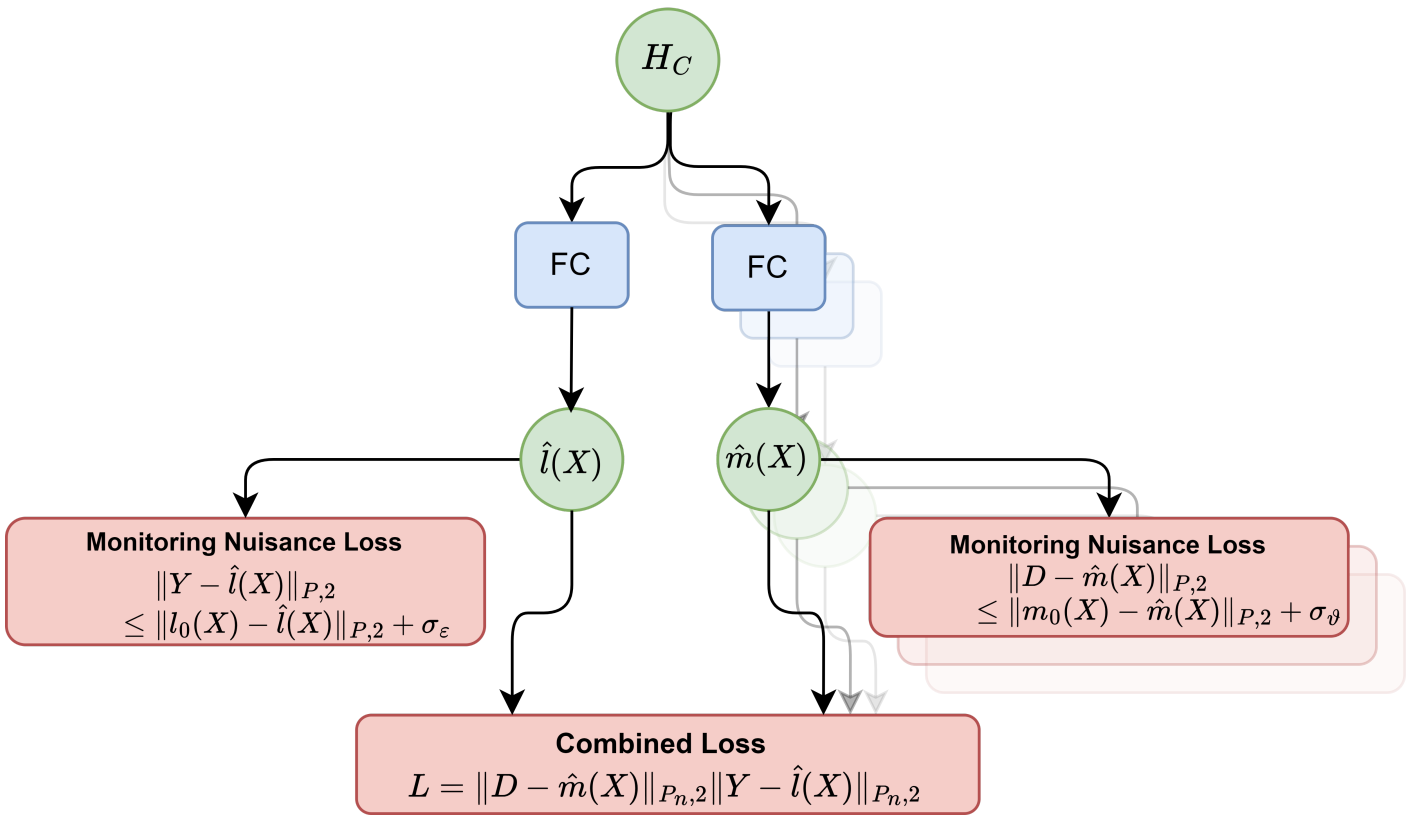
# DoubleMLDeep

# DoubleMLDeep - Module Structure




High-Level PLR Model Architecture. Both nuisance components are trained simultaneously with a combined loss.

# DoubleMLDeep - Module Structure




Nuisance Estimation with Combined Loss in case of multiple treatments with nuisance loss monitoring.

- It is possible to use binary treatment variables $D \in \{0, 1\}$ by adjusting the loss function:

$$L = L_{log}(D, \hat{m}(X))\|Y - \hat{l}\,(X)\|_{P_n,2}$$

# Application

# Amazon Toys Dataset

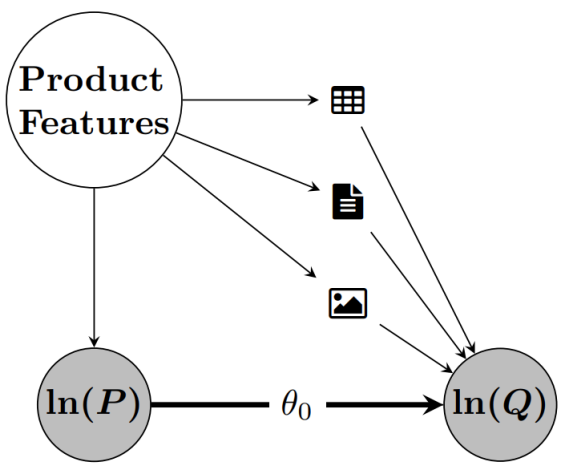
- Public Data from the Amazon sales platform from category toys and games for subcategories vehicles, cars and trucks.
- Using the Sales Rank as a proxy for quantities, as shown in Bajari et al. (2023).

| Variable | Description |
|---|---|
| Sales Rank | Sales rank as weighted mean for the last 30 days |
| Price | Price as weighted mean for the last 30 days |
| Tabular | Tabular features as reviews, ratings, etc. |
| Text | Combination of Title, Category, Description, etc. |
| Image | Image of the product |

# Model

- Run a simple log-log regression model
- Images and Text block backdoor path from price to demand (or sales rank)

$$\ln(Q) = \theta_0 \ln(P) + g_0(X) + \epsilon$$



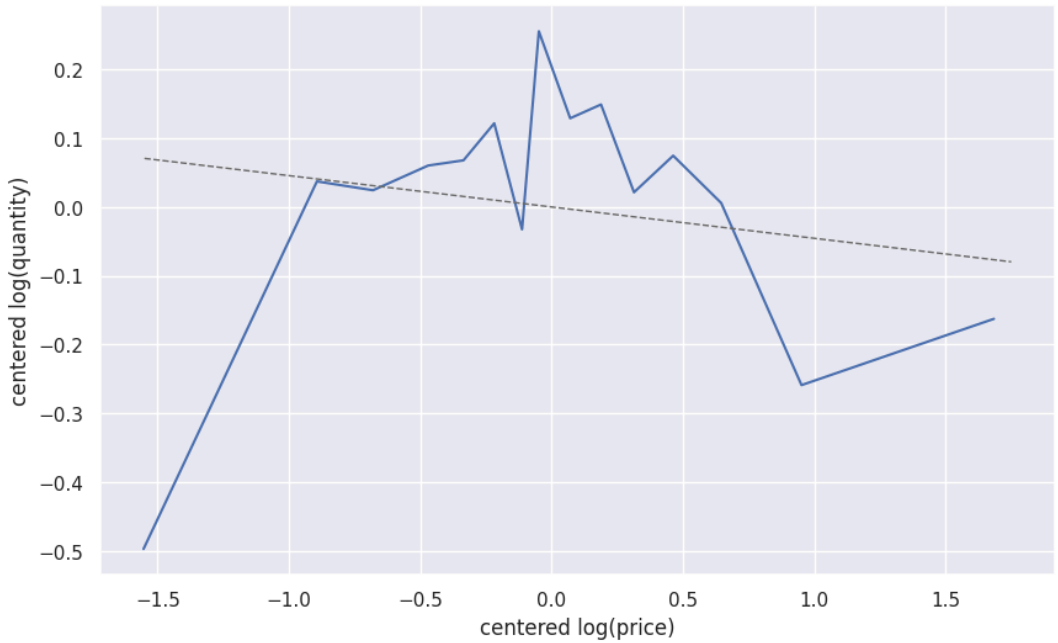
# Baseline OLS Model

- Baseline estimate with tabular covariates $X_{\text{tab}}$
- OLS:

$$\ln(Q) = \theta_0 \ln(P) + \beta^T X_{\text{tab}} + \epsilon$$

- $R^2 = 0.330$

| 2.5 % | $\hat{\theta}$ | 97.5 % |
|---|---|---|
| -0.072 | -0.046 | -0.019 |

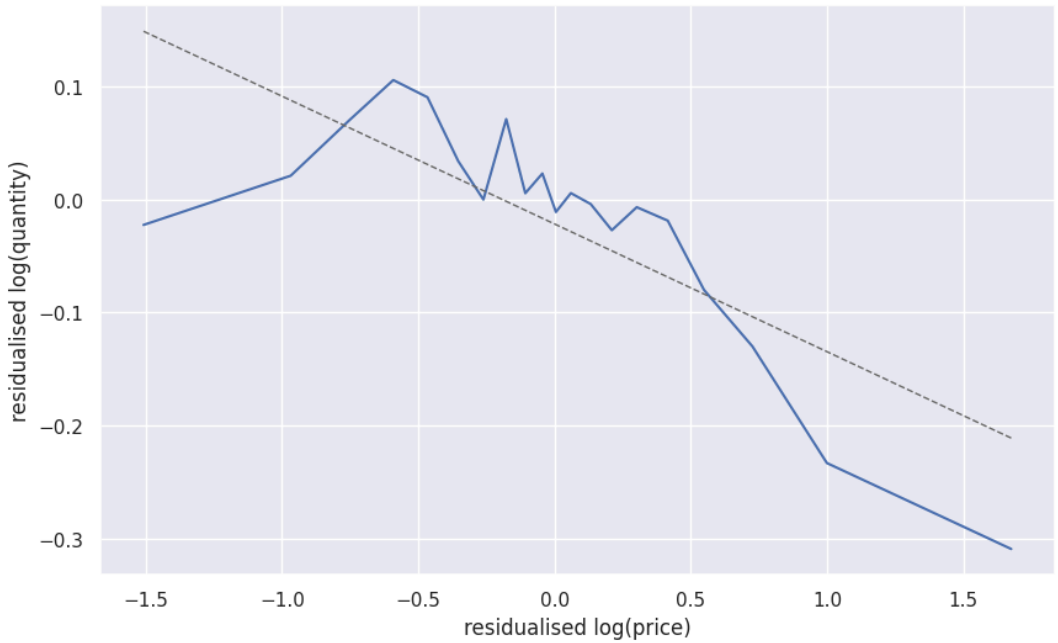



Centered logarithmic negative sales rank over centered logarithmic price as binned plot

# Baseline DML Model

- Estimate with tabular data as covariates $X_{\text{tab}}$ in `DoubleMLPLR` model with `RandomForest` regressors
  - $l_0(X_{\text{tab}}) := \mathbb{E}[\ln(Q)|X_{\text{tab}}]$
    - $R^2_{l_0} = 0.5986$
  - $m_0(X_{\text{tab}}) := \mathbb{E}[\ln(P)|X_{\text{tab}}]$
    - $R^2_{m_0} = 0.1884$

| **2.5 %** | $\theta$ | **97.5 %** |
|---|---|---|
| -0.132 | -0.1098 | -0.080 |




Residualised logarithmic negative sales rank over residualised logarithmic price as binned plot

# Deep Learning Models

- The `DoubleMLPLRDeep` was trained using sets of covariates augmented with text, image, or both modalities:

| Used Confounders |
|---|
| $X = (X_{txt}, X_{tab})$ |
| $X = (X_{img}, X_{tab})$ |
| $X = (X_{txt}, X_{img}, X_{tab})$ |

- With each set of covariates $X$, 5 runs are evaluated.
- The models were trained over 10 epochs with early stopping callback.
- The data was split into 80% training and 20% validation data.

$\Rightarrow$ The aim is to emphasize the benefits of utilizing unstructured data.

# (First) Results

| Model[1] | Covariates | $R^2_{l_0}$ | $R^2_{m_0}$ | $\hat{\theta}_0$ |
|---|---|---|---|---|
| `OLS` | $X = X_{tab}$ | $0.3300$ | - | $-0.0455$ |
| `DoubleMLPLR` | $X = X_{tab}$ | $0.5986$ | $0.1884$ | $-0.1098$ |
| `DoubleMLPLRDeep` | $X = (X_{txt}, X_{tab})$ | $0.5574 \pm 0.025$ | $0.6188 \pm 0.015$ | $-0.2137 \pm 0.053$ |
| `DoubleMLPLRDeep` | $X = (X_{img}, X_{tab})$ | $0.4518 \pm 0.029$ | $0.5085 \pm 0.028$ | $-0.2721 \pm 0.038$ |
| `DoubleMLPLRDeep` | $X = (X_{txt}, X_{img}, X_{tab})$ | $0.5829 \pm 0.015$ | $0.6532 \pm 0.018$ | $-0.2794 \pm 0.044$ |

- Estimates $\hat{\theta}$ become more negative with the inclusion of unstructured data
- The $R^2$-scores of the nuisance functions improve with the inclusion of unstructured data
- Semi-synthetic data can illustrate the potential of unstructured data (see Appendix)

# Future Directions

- Explore settings with multiple treatment variables.
- Investigate heterogeneous treatment effects.
- Apply the DoubleMLDeep framework in Difference-in-Differences (DiD) and Instrumental Variables (IV) settings.
- Conduct extensive simulation studies to test the robustness and accuracy of the DoubleMLDeep framework under various conditions.

$\Rightarrow$ **You can find more information in our paper: Klaassen et al. (2024)**

# Thank you for your attention!




University of Hamburg

https://www.bwl.uni-hamburg.de/en/statistik/allgemein.html




Massachusetts Institute of Technology

https://stat.mit.edu/about/




Economic AI

https://economicai.com/




DoubleML

https://docs.doubleml.org/stable/index.html

# References

# References


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# Appendix: Double Machine Learning

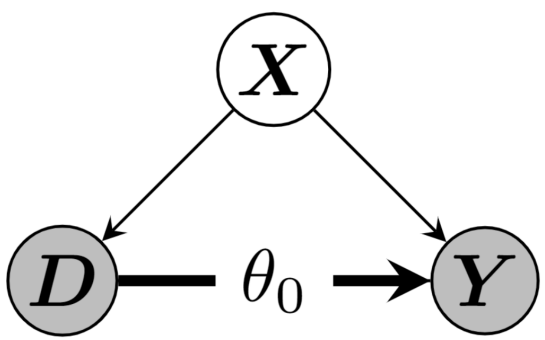
# Motivating Example - PLR

- Revisit the PLR model from the beginning:

$$
\begin{aligned}
Y &= \theta_0 D + g_0(X) + \varepsilon, & \mathbb{E}[\varepsilon | X, D] &= 0 \\
D &= m_0(X) + \vartheta, & \mathbb{E}[\vartheta | X] &= 0
\end{aligned}
$$

with

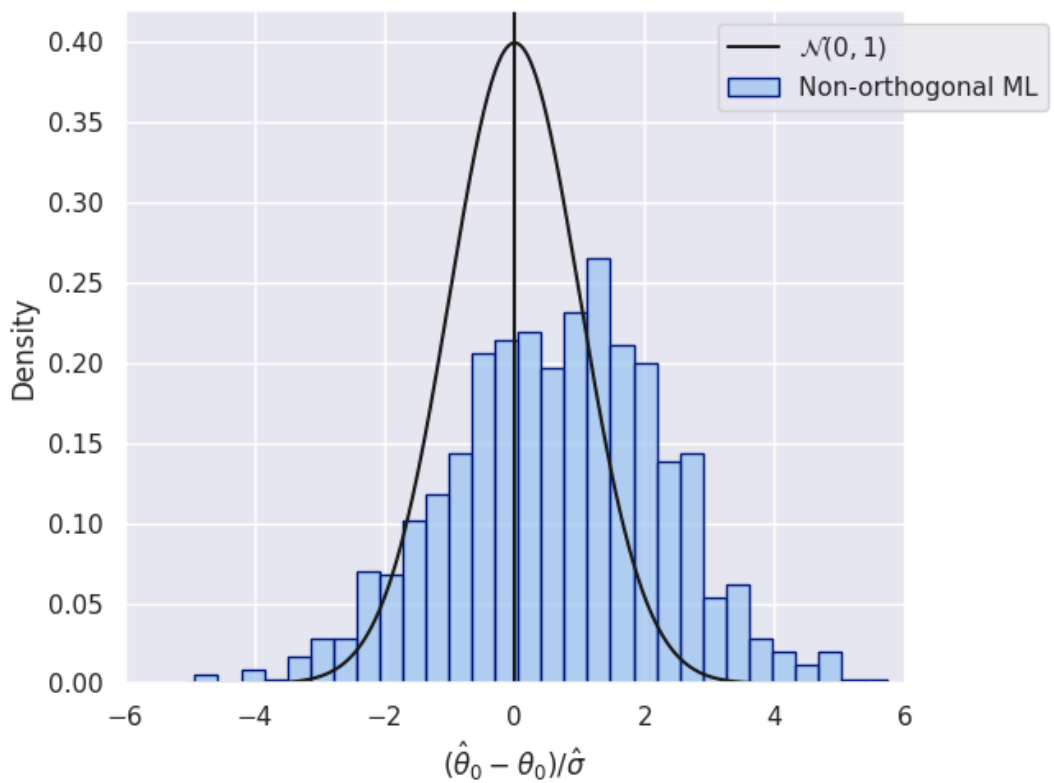
- $Y$ - outcome variable
- $D$ - policy/treatment variable
- $X = (X_1, \ldots X_p)^T$ - vector of additional controls
- $\varepsilon, \vartheta$ - stochastic errors




DAG for PLR Model

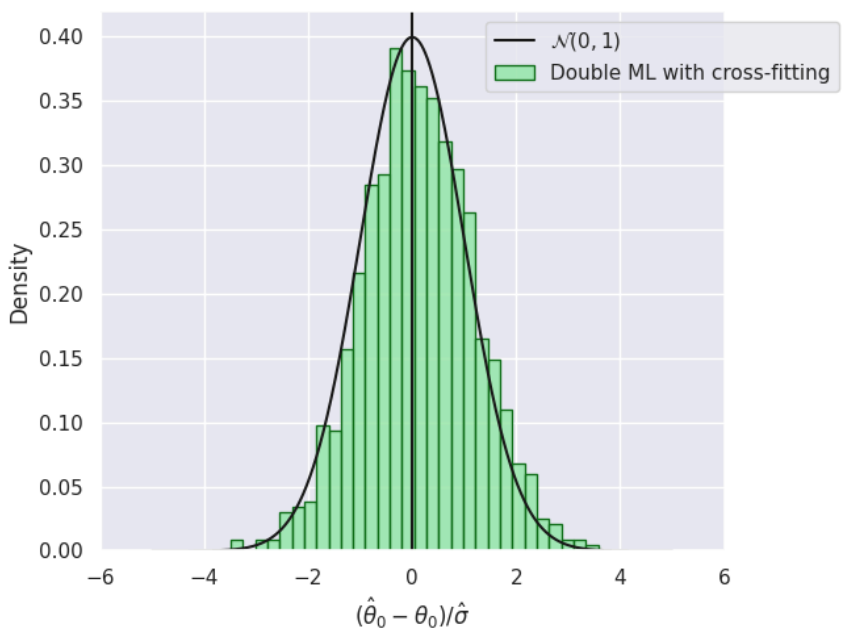
# Motivating Example - Regularization Bias

- What if we simply plug-in ML predictions $\hat{g}(X)$ for $g_0(X)$ into $Y = \theta_0 D + g_0(X) + \varepsilon$?



# Motivating Example - Orthogonalization

- *Frisch-Waugh-Lovell* style approach: $\theta_0$ can be consistently estimated by **partialling out** $X$, i.e,
  1. Predict $Y$ and $D$ by $\mathbb{E}[Y|X]$ and $\mathbb{E}[D|X]$, obtained using ML methods
  2. Residualize $\tilde{Y} = Y - \mathbb{E}[Y|X]$ and $\tilde{D} = D - \mathbb{E}[D|X]$
  3. Regress $\tilde{Y}$ on $\tilde{D}$ to obtain $\hat{\theta}$



# Double/Debiased Machine Learning (DML)

- **Neyman Orthogonality**

$$\partial_\eta \mathbb{E}[\psi(W; \theta_0, \eta)]|_{\eta=\eta_0} = 0$$

ensures that the moment condition identifying $\theta_0$ is **insensitive to small pertubations** of the nuisance function $\eta$ around $\eta_0$

- Assumptions for the nuisance elements (see Chernozhukov et al. (2018a))

$$\|\hat{m} - m_0\|_{P,2} \times \left(\|\hat{m} - m_0\|_{P,2} + \|\hat{\ell} - \ell_0\|_{P,2}\right) \leq \delta_N N^{-1/2}$$

- Under some regularity conditions the estimator $\hat{\theta}$ concentrates in $1/\sqrt{N}$-neighborhood of $\theta_0$ and

$$\sqrt{N}(\hat{\theta} - \theta_0) \to \mathcal{N}(0, \sigma^2)$$

$\Rightarrow$ simple approach to control for confounders, *if they can be used for ML models!*

# Neyman Orthogonality

The naive approach minimizes the following MSE

$$min_\theta \mathbb{E}[(Y - D\theta_0 - g_0(X))^2]$$

This implies the following moment equation

$$\mathbb{E}[\underbrace{(Y - D\theta_0 - g_0(X))D}_{=:\psi(W,\theta_0,\eta)}] = 0$$

Whereas for the partialling-out approach minimizes

$$min_\theta \mathbb{E}\left[\left(Y - \mathbb{E}[Y|X] - (D - \mathbb{E}[D|X])\theta\right)^2\right]$$

which implies

$$\mathbb{E}\Big[\underbrace{\left(Y - \mathbb{E}[Y|X] - (D - \mathbb{E}[D|X])\theta\right)(D - \mathbb{E}[D|X])}_{=:\psi(W,\theta_0,\eta)}\Big] = 0$$

# Neyman Orthogonality

**Naive approach**

$$\psi(W, \theta_0, \eta) = (Y - D\theta_0 - g_0(X))D$$

Regression adjustment score

$$\begin{aligned} \eta &= g(X), \\ \eta_0 &= g_0(X). \end{aligned}$$

**FWL partialling out**

$$\begin{aligned} \psi(W, \theta_0, \eta_0) = &\Big((Y - E[Y|X]) \\ &- (D - E[D|X])\theta_0\Big) \\ &(D - E[D|X]) \end{aligned}$$

Neyman-orthogonal score (Frisch-Waugh-Lovell)

$$\begin{aligned} \eta &= (\ell(X), m(X)), \\ \eta_0 &= (\ell_0(X), m_0(X)), \\ &= (\mathbb{E}[Y \mid X], \mathbb{E}[D \mid X]). \end{aligned}$$

# DML Key Ingredients

## 1. Neyman Orthogonality

- Inference is based on a moment condition that satisfies the **Neyman orthogonality condition** $\psi(W; \theta, \eta)$

$$E[\psi(W; \theta_0, \eta_0)] = 0,$$

- where $W := (Y, D, X, Z)$ and with $\theta_0$ being the unique solution that obeys the **Neyman orthogonality condition**

$$\partial_\eta \mathbb{E}[\psi(W; \theta_0, \eta)]|_{\eta=\eta_0} = 0.$$

- $\partial_\eta$ denotes the pathwise (Gateaux) derivative operator

# DML Key Ingredients

### 1. Neyman Orthogonality

- **Neyman orthogonality** ensures that the **moment condition** identifying $\theta_0$ is **insensitive to small pertubations** of the nuisance function $\eta$ around $\eta_0$
- Using a Neyman-orthogonal score **eliminates the first order biases** arising from the replacement of $\eta_0$ with a ML estimator $\hat{\eta}_0$
- PLR example: Partialling-out score function

$$\psi(\cdot) = (Y - E[Y|X] - \theta(D - E[D|X]))(D - E[D|X])$$

# DML Key Ingredients

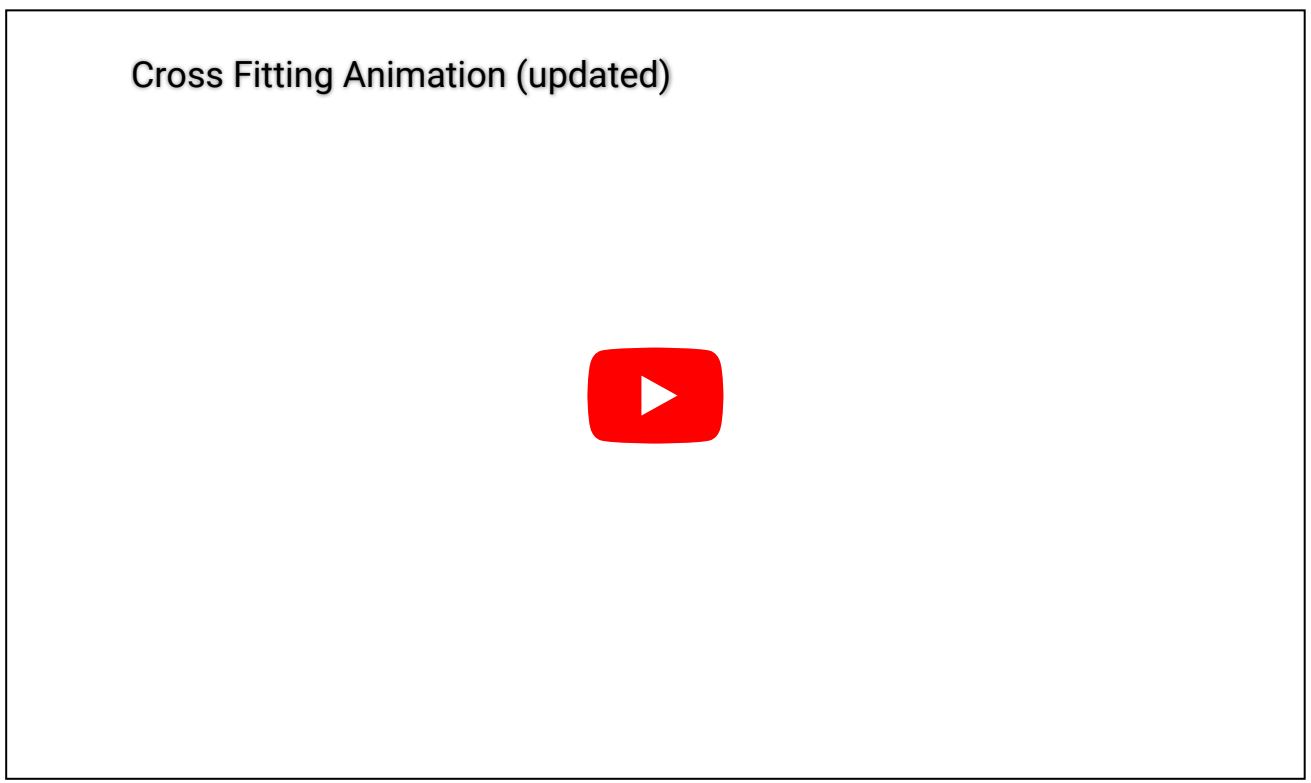
### 2. High-Quality Machine Learning Estimators

- The nuisance parameters are estimated with high-quality (fast-enough converging) machine learning methods.
- Different structural assumptions on $\eta_0$ lead to the use of different machine-learning tools for estimating $\eta_0$ Chernozhukov et al. (2018a) (Section 3)
- Rate requirements depend on the causal model and orthogonal score, e.g. (see Chernozhukov et al. (2018a)),
  - PLR, partialling out:
    $\|\hat{m}_0 - m_0\|_{P,2} \times \left(\|\hat{m}_0 - m_0\|_{P,2} + \|\hat{\ell}_0 - \ell_0\|_{P,2}\right) \leq \delta_N N^{-1/2}$
  - IRM/DR score, ATE: $\|\hat{m}_0 - m_0\|_{P,2} \times \|\hat{\ell}_0 - \ell_0\|_{P,2} \leq \delta_N N^{-1/2}$

# DML Key Ingredients

### 3. Sample Splitting

- To avoid the biases arising from overfitting, a form of **sample splitting** is used at the stage of producing the estimator of the main parameter $\theta_0$.
- Efficiency gains by using cross-fitting (swapping roles of samples for train / hold-out)



# DML Key Ingredients

**Main result in Chernozhukov et al. (2018a)**

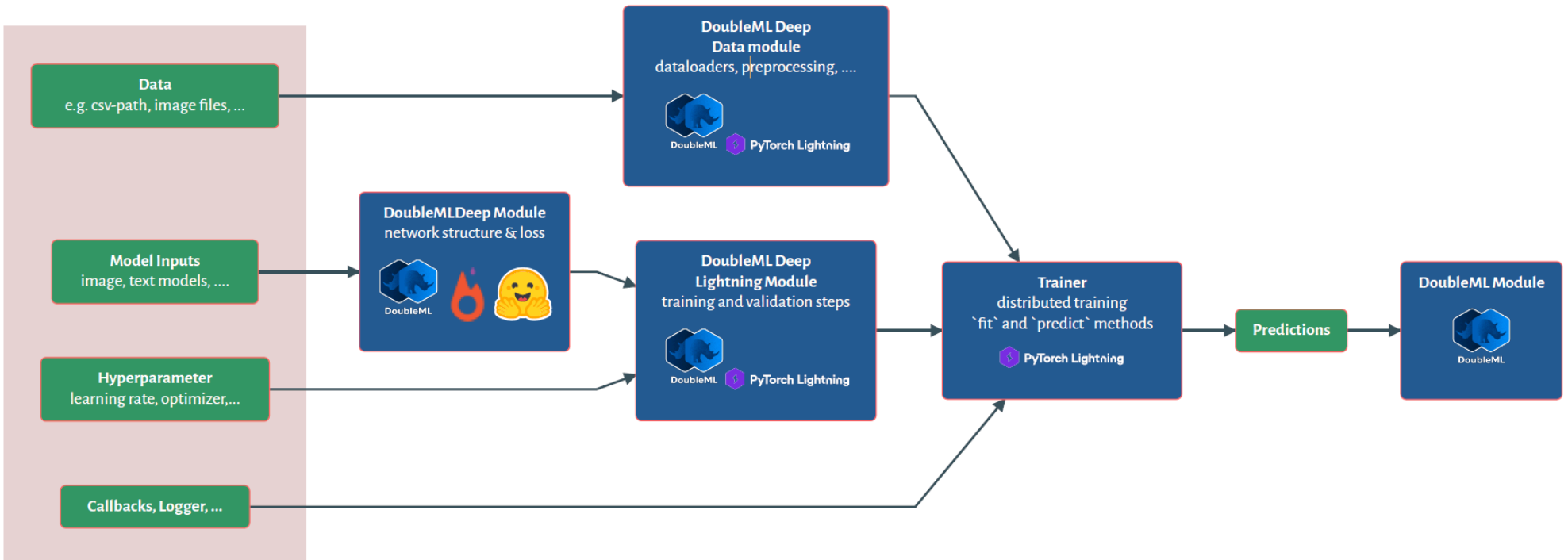
> There exist regularity conditions, such that the DML estimator $\tilde{\theta}_0$ concentrates in a $1/\sqrt{N}$-neighborhood of $\theta_0$ and the sampling error is approximately
>
> $$\sqrt{N}(\tilde{\theta}_0 - \theta_0) \sim N(0, \sigma^2),$$
>
> with
>
> $$\sigma^2 := J_0^{-2}\mathbb{E}(\psi^2(W; \theta_0, \eta_0)),$$
> $$J_0 = \mathbb{E}(\psi_a(W; \eta_0)).$$

- See this example based on Chernozhukov et al. (2018a)

# DoubleMLDeep - More Details

- Use pretrained models for unstructured data e.g. Bert and Beit
- Basic models for tabular data seem sufficient
- Monitor the nuisance losses
  - $\|Y - \hat{l}(X)\|_{P,2} \leq \|l_0(X) - \hat{l}(X)\|_{P,2} + \sigma_\epsilon$
  - $\|D - \hat{m}(X)\|_{P,2} \leq \|m_0(X) - \hat{m}(X)\|_{P,2} + \sigma_\vartheta$

# DoubleMLDeep - Workflow




High-Level Workflow of *DoubleMLDeep*

# Appendix: Application (Amazon Toys Data)

# Amazon Toys Dataset

### Continous Variables

- Reviews: Rating
- Reviews: Review Count
- New Offer Count: Current
- Count of retrieved live offers: New, FBA
- Count of retrieved live offers: New, FBM

### Categorical Variables

- Lightning Deals: Upcoming Deal
- Buy Box: Is FBA

# Deep Learning Models

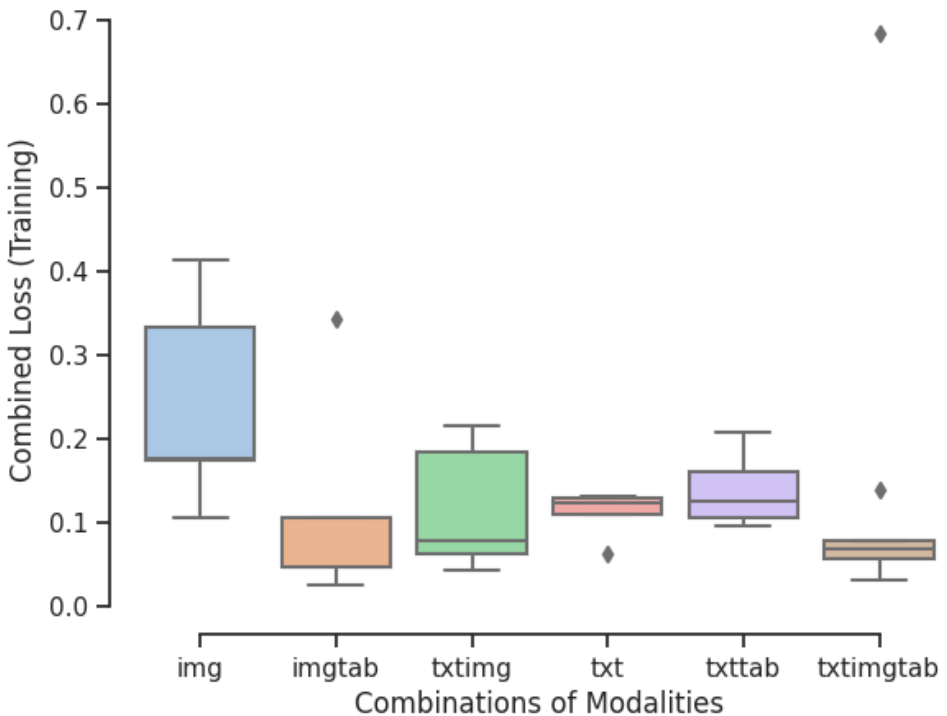
- In this study, a variety of input data combinations were used to train `DoubleMLPLRDeep` models [1]:

| Key | Used Confounders |
|---|---|
| `img` | $X = (X_{img})$ |
| `imgtab` | $X = (X_{img}, X_{tab})$ |
| `txtimg` | $X = (X_{txt}, X_{img})$ |
| `txt` | $X = (X_{txt})$ |
| `txttab` | $X = (X_{txt}, X_{tab})$ |
| `txtimgtab` | $X = (X_{txt}, X_{img}, X_{tab})$ |

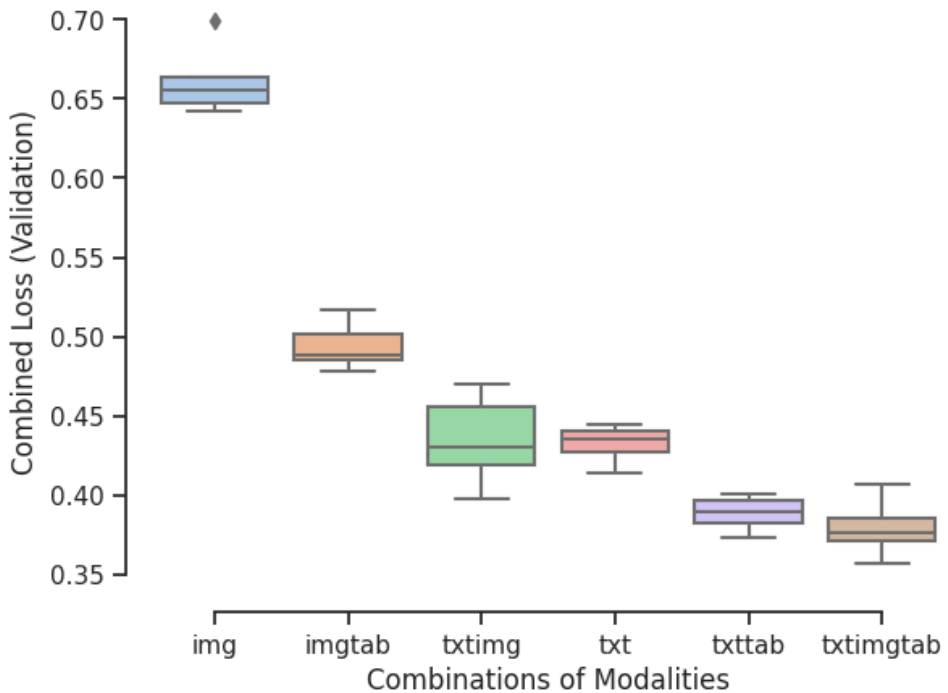
$\Rightarrow$ The aim is to emphasize the benefits of utilizing unstructured data.

# Deep Learning Models: RMSE-Scores

- Combined RMSE Score (Training)

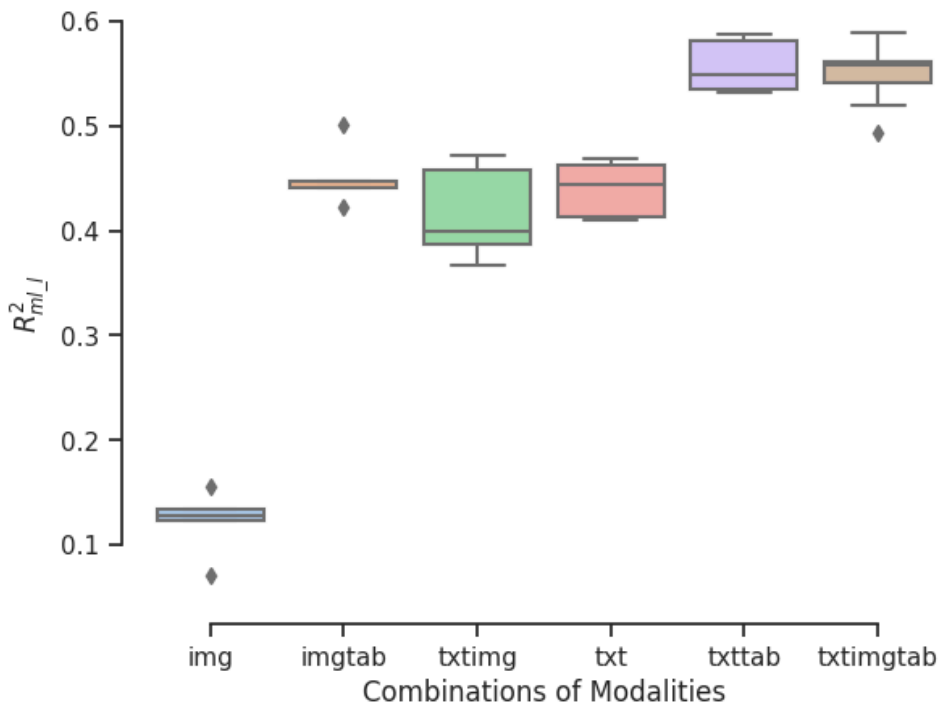



- Combined RMSE Score (Validation)

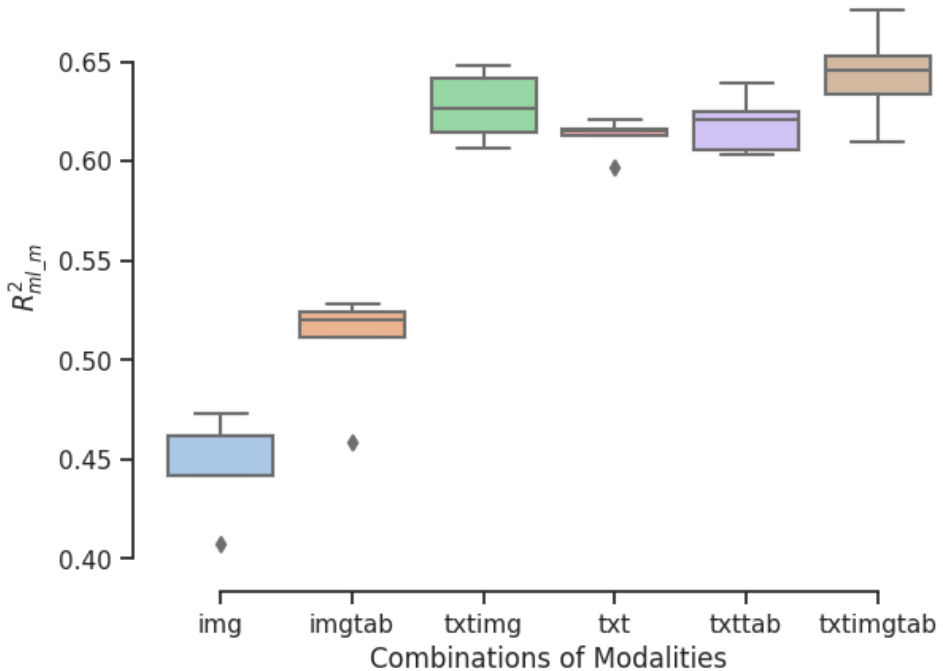


# Deep Learning Models: $R^2$-Scores

- $R^2$ of log(Quantity) on Validation Set




- $R^2$ of log(Price) on Validation Set

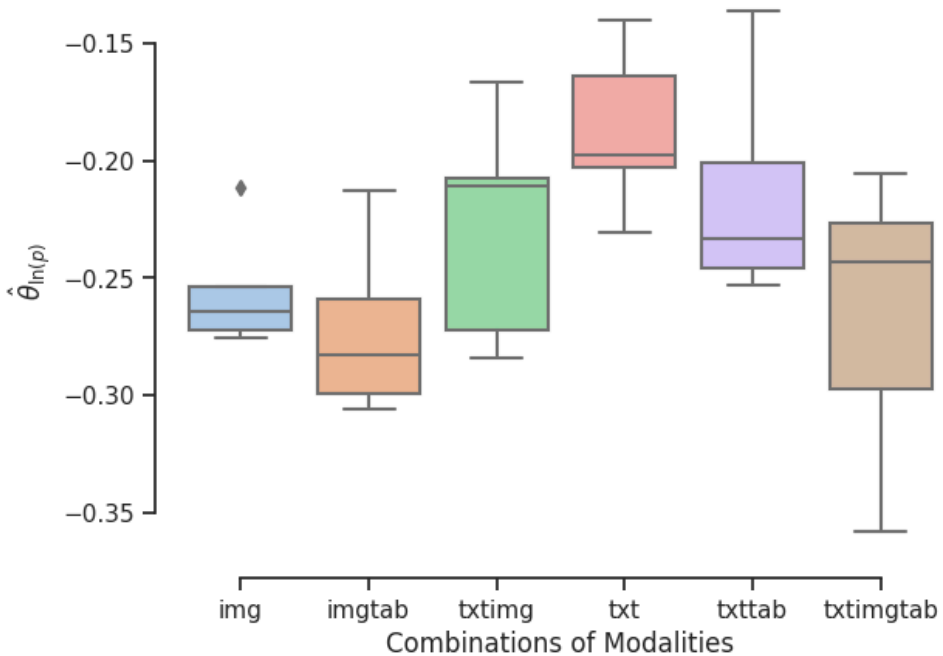


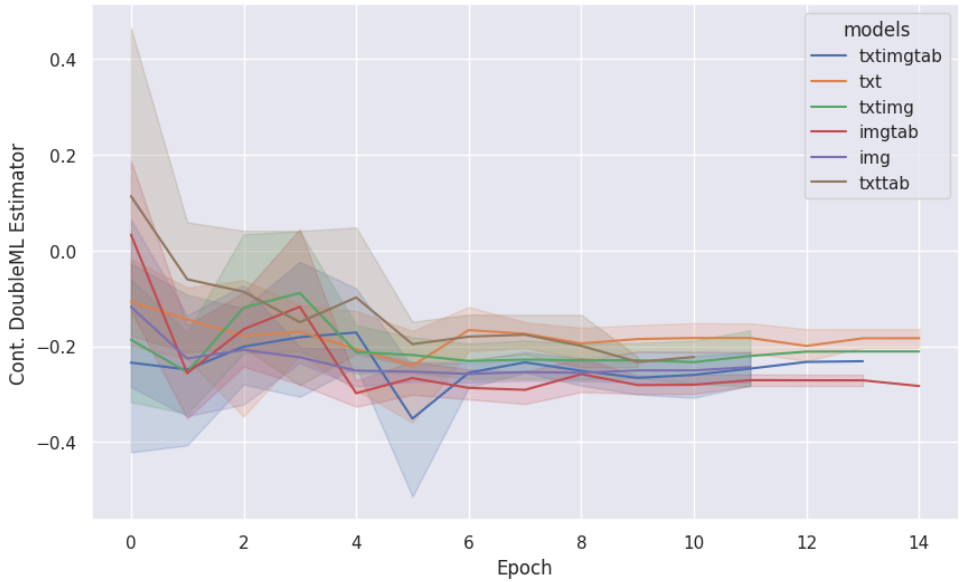
# Scatterplots with OLS Regression of Tabular Variables and $\ln(Q)$



# (First) Results

- Using different combinations of tabular, image and text data we obtain the following estimates




Continous Estimates from `DoubleMLPLR` on Validation Set during the Training Process

# Appendix: Semi-Synthetic Data

# Original Datasets

- We generate a semi-synthetic dataset with a known treatment effect parameter
- To generate credible confounding wil be based on the **labels** or **outcomes** of the corresponding supervised learning task
- We use the following datasets:

| **Modality** | **Dataset** | **Target** $\tilde{X}$ | **Control** $X$ |
|---|---|---|---|
| Tabular | DIAMONDS<br>Wickham (2016) | $\log(\text{Price})$ | Carat, Cut, Color, Clarity, ... |
| Text | IMDB<br>Maas et al. (2011) | Sentiment | Review Text |
| Image | CIFAR-10<br>Krizhevsky (2009) | Label | Image |

# Data Generating Process

We generate a semi-synthetic dataset according to the underlying PLR model

$$
\begin{aligned}
Y &= \theta_0 D + \tilde{g}_0(\tilde{X}) + \varepsilon, \\
D &= \tilde{m}_0(\tilde{X}) + \vartheta,
\end{aligned}
$$

where $\tilde{X} = (\tilde{X}_{\text{tab}}, \tilde{X}_{\text{txt}}, \tilde{X}_{\text{img}})$ with the following additive structure

$$
\begin{aligned}
\tilde{g}_0(\tilde{X}) &= \sum_{\text{mod} \in \{\text{tab,txt,img}\}} \tilde{g}_{\text{mod}}(\tilde{X}_{\text{mod}}) \\
\tilde{m}_0(\tilde{X}) &= \sum_{\text{mod} \in \{\text{tab,txt,img}\}} \tilde{m}_{\text{mod}}(\tilde{X}_{\text{mod}})
\end{aligned}
$$

and $\varepsilon, \vartheta \sim \mathcal{N}(0, 1)$.

# Data Generating Process

- The effect on the outcome $Y$ is generated via a standardized version of target variable to balance the confounding impact of all modalities:

$$\tilde{g}_{\text{mod}}(\tilde{X}_{\text{mod}}) = \frac{\tilde{X}_{\text{mod}} - \mathbb{E}[\tilde{X}_{\text{mod}}]}{\sigma_{\tilde{X}_{\text{mod}}}}, \quad \text{mod} \in \{\text{tab}, \text{txt}, \text{img}\}$$

- Further, to ensure a strong confounding, the impact on the treatment $D$ is defined via:

$$\tilde{m}_{\text{mod}}(\tilde{X}_{\text{mod}}) = -\tilde{g}_{\text{mod}}(\tilde{X}_{\text{mod}}), \quad \text{mod} \in \{\text{tab}, \text{txt}, \text{img}\}$$

- The treatment effect is set to $\theta_0 = 0.5$ with $n = 50.000$ samples in the dataset
- Both $\tilde{g}_0(X)$ and $\tilde{m}_0(X)$ are rescaled to ensure a signal-to-noise ratio of $2$ for $Y$ and $D$ (given unit variances of the error terms)

# Challenges

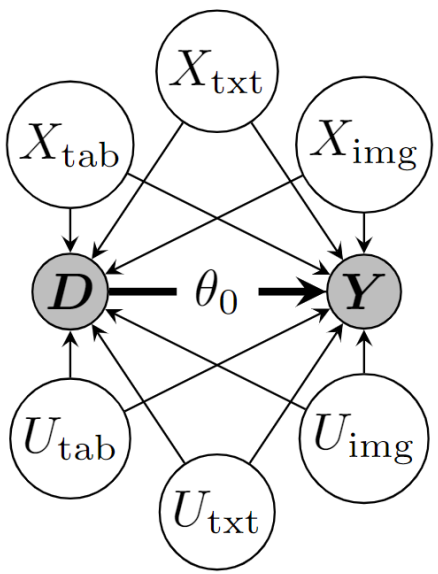
- Dependency on the target of the modality $\tilde{X}_{\text{mod}}$
  - Might not be fully explained by the corresponding features $X_{\text{mod}}$
  - E.g. price of diamonds can not be fully explained by the carat, cut, color, etc.
- Consequently, the estimate $\hat{\theta}$ might only be able to account for the part of confounding which can be explained by the input features:

$$\tilde{X}_{\text{mod}} = \mathbb{E}[\tilde{X}_{\text{mod}} | X_{\text{mod}}] + U_{\text{mod}},$$

  where $U_{\text{mod}}$ can not be controlled for

# Challenges

- Due to the negative sign and the additive structure, the confounding effect will ensure that higher outcomes $Y$ occur with lower treatment values $D$, creating a **negative bias**
- The independence of all three original datasets and the additive negative confounding results in a negative bias even if we only control for a subset of confounding factors




DAG for the semi-synthetic dataset. The confounding via the features $X = (X_{\text{tab}}, X_{\text{txt}}, X_{\text{img}})$ can be adjusted for, whereas the unexplained/noise parts $U = (U_{\text{tab}}, U_{\text{txt}}, U_{\text{img}})$ are unobserved.

# Bounds for the estimation of $\theta_0$

- Since all modalities contribute a negative bias, the semi-synthetic dataset can be used as a benchmark with an oracle upper bound of an effect estimate of

$$\theta_0 = 0.5$$

- For the lower bound, one can not account for confounding at all and evaluate a basic ordinary least squares model with outcome $Y$ on the treatment variable $D$ (excluding all confounding variables)

$$\hat{\theta}_{\text{OLS}} = -0.4594$$

# Bounds for the estimation of $\eta_0$

- To access the predictive performance of the nuisance models, we can rely on oracle predictions of

$$\begin{aligned}\tilde{m}_0(\tilde{X}) &:= \mathbb{E}[D|\tilde{X}] \\ \tilde{l}_0(\tilde{X}) &:= \mathbb{E}[Y|\tilde{X}] = \theta_0 \tilde{m}_0(\tilde{X}) + \tilde{g}_0(\tilde{X})\end{aligned}$$

- Evaluating the oracle predictions $\tilde{m}_0(\tilde{X})$ and $\tilde{l}_0(\tilde{X})$ results in the following upper bounds for the performance of the nuisance estimators

$$\begin{aligned}R^2(D, \tilde{m}_0(\tilde{X})) &= 0.6713 \\ R^2(Y, \tilde{l}_0(\tilde{X})) &= 0.5845\end{aligned}$$

  on the whole dataset of $n = 50.000$ observations

# Bounds for the estimation of $\eta_0$

- In order to compare the predictive performance of different models, a relative $r^2$-score with respect to the upper bound is defined as

$$0 \leq r^2(D, \hat{m}) := \frac{R^2(D, \hat{m}(X))}{R^2(D, \tilde{m}_0(\tilde{X}))} \leq 1$$

$$0 \leq r^2(Y, \hat{l}) := \frac{R^2(Y, \hat{l}(X))}{R^2(Y, \tilde{l}_0(\tilde{X}))} \leq 1$$

# Models

- **Baseline Model**
  - Standard DML approach, relying only on tabular data $X_{\text{tab}}$
  - Based on the LightGBM package for estimation of the nuisance elements
- **Deep Model**
  - Uses the out-of-sample predictions of $\hat{m}(X)$ and $\hat{l}(X)$ generated from the model
  - Based on the RoBERTa Model for text modality, a VIT Model for images and a SAINT model for the tabular data
- **Embedding Model**
  - Generated embedding $H_E$ is used together with the tabular features $X_{\text{tab}}$ as input for LightGBM
  - Based on the RoBERTa Model for text modality, a VIT Model for images and a SAINT model for the tabular data

# Simulation Results

Results of Simulation Study. Reported: mean ± sd. over five random train-test splits

| | **Baseline** | **Embedding** | **Deep** |
|---|---|---|---|
| $r^2(Y, \hat{l}_0)$ | $0.31 \pm 0.01$ | $0.87 \pm 0.02$ | $\mathbf{0.90 \pm 0.01}$ |
| $r^2(D, \hat{m}_0)$ | $0.31 \pm 0.01$ | $0.87 \pm 0.02$ | $\mathbf{0.90 \pm 0.01}$ |
| $\hat{\theta}$ | $-0.32 \pm 0.01$ | $\mathbf{0.28 \pm 0.01}$ | $0.27 \pm 0.01$ |

*Higher = better (best in bold)*

# Simulation Results - Performance of $\hat{\theta}$




Boxplots of $\hat{\theta}$. The Embedding Model and Deep Model have DoubleML Deep estimates. This indicates a stable and information-rich